



Designer probiotics for the prevention and treatment of human diseases

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Various studies have shown the beneficial effects of probiotics in humans. The use of synthetic biology to engineer programmable probiotics that specifically targets cancer, infectious agents, or other metabolic diseases has gained much interest since the last decade. Developments made in synthetic probiotics as therapeutics within the last three years will be discussed in this review.

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Introduction

The human body is long known to be an ecosystem on its own, made up of not only human cells but also microbes, otherwise known as commensal microflora. They are found on the epithelial surfaces, exposed to the external environment, and are most abundant in the gastrointestinal tract (GIT) [1]. The body provides a stable environment for the microflora, while the microbes protect the body by defending against colonization of pathogens as well as stimulating the host's immune system. Commensal microbes are also known to ferment fibers to provide prebiotics [2] as well as produce folate [3], both of which are important in improving the host's health.

It remains unknown whether dysbiosis, an imbalance in the composition of the microbes, is the cause or result of various metabolic and immune diseases. Although probiotics are commonly used as an attempt to restore this imbalance, the optimal therapeutic strain remains to be determined [4]. Probiotics are non-pathogenic bacteria or

yeast that can survive the harsh environment in the GIT conferring health benefits to the host [5].

The emergence of synthetic biology allows engineering of commensal microbes and probiotics to perform novel therapeutic functions. Over the years, scientists have engineered probiotics to target diseases such as colitis [6], diabetes [7], as well as HIV infections [8]. Here, we will review recent developments in synthetic probiotic therapeutics that not only address metabolic and infectious diseases, but also immune diseases, allergies as well as cancers (Figure 1).

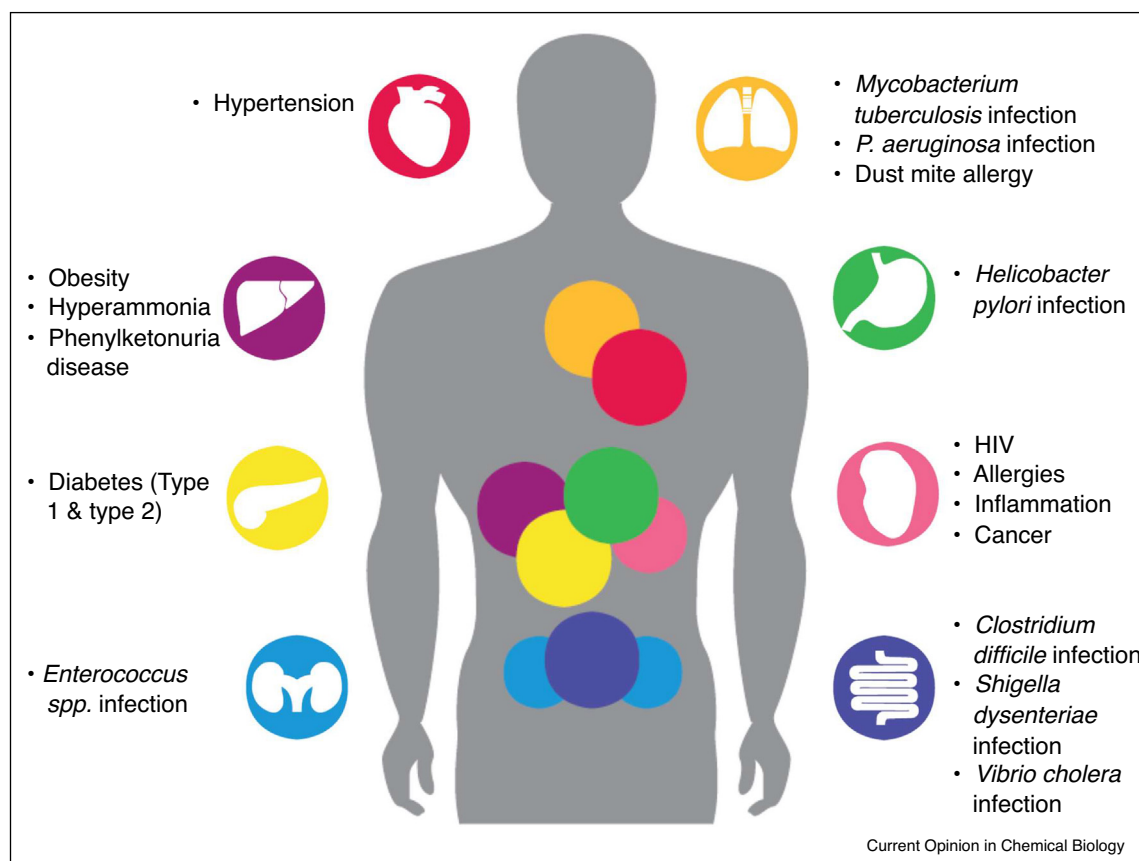
Engineering probiotics for treatment of inherited and acquired metabolic disorders

Metabolic disorders are costly to treat and require strict compliance that affects patients' lifestyle and survival rates. Engineering microbes that can reside in patients' gastrointestinal tract whilst treating and alleviating patient's symptoms provides long lasting benefits.

With more than 1 billion people worldwide predicted to live with hypertension by 2025, there is an urgent need to address this health issue. Angiotensin-converting enzyme inhibitory peptides (ACEIPs) help blood vessels to relax and reduce water reabsorption by the kidneys, thereby lowering blood pressure. *Lactobacillus plantarum* NC8 strain modified to deliver ACEIPs, where its coding sequences from tuna frame protein (TFP) and yellow fin sole frame protein (YFP) are joined by an arginine linker, was observed to effectively decrease systolic blood pressure, endothelin and angiotensin II levels in spontaneously hypertensive rats [9]. While traditional ACEIPs were short-acting with the anti-hypertension effects lasting for a day, administration of modified *L. plantarum* conferred anti-hypertensive properties that lasted for an additional 10 days after the initial 24-day treatment period with no observable side effects *in vivo*.

Diabetes mellitus type 1 (DM1) is an inherited disease where insulin-producing pancreatic β cells are destroyed by autoreactive T cells, eventually causing patients to rely on insulin injections to manage blood glucose levels. Engineered *Lactococcus lactis* NZ9000 expressing fusion protein HSP65-6P277 was found to impede DM1 onset in non-obese diabetic (NOD) mice with markedly improved glucose tolerance and reduced insulinitis [10].

Figure 1



A schematic of the various target organs and diseases of engineered microbial therapeutics. Other than the commonly associated gastrointestinal tract, these therapeutics can be used in almost all organs and against a wide variety of diseases. Probiotic therapeutics for their respective diseases named above are discussed in more detail in the text.

Besides delaying DM1 onset, engineered bacteria can also help manage Diabetes mellitus type 2 (DM2). Glucagon-like peptide-1 (GLP-1), an incretin-derived hormone, improves pancreatic function but is limited by its short half-life. Nevertheless, its potential therapeutic application can be achieved when commensal bacteria such as *Bifidobacterium longum* is engineered to express and secrete biologically active GLP-1 directly in the colon in the form of penetratin-GLP-1 fusion protein [11]. The inactive full-length form of GLP-1 (1–37) expressed in *Lactobacillus gasseri* ATCC 33323 was shown to help convert intestinal epithelial cells into insulin-secreting cells in the upper intestine of diabetic rats [7]. When fed daily with the engineered probiotics, insulin production from the converted cells covered 25–33% of the insulin capacity. Similarly, exendin-4 peptide, a GLP-1 receptor agonist expressed in *Lactobacillus paracasei* reportedly enhanced insulin secretion in β cells, providing yet another sustained and non-chemical alternative to diabetic patients [12].

Other than hypertension and diabetes, obesity is another metabolic disease that affects an increasing number of

individuals. Engineered *Escherichia coli* Nissle 1917 (EcN) producing *N*-acylphosphatidylethanolamines (NAPEs) prevented obesity in mice [13], while those engineered to produce redox factor pyrroloquinoline quinone (*pqq*) together with fructose dehydrogenase (*pqq-fdh*), or with glucose facilitator protein and mannitol-2-dehydrogenase (*pqq-glf-mtlK*) helped treat fructose induced hepatic steatosis in mice [14]. On the other hand, mice with iron deficiency that received combined treatment of EcN (*pqq-glf-mtlK*) and fructose showed a marked improvement in transferrin-bound iron levels without the undesired metabolic syndromes caused by fructose consumption [15].

Engineering probiotics for treatment and prevention of infectious diseases

With the emergence of multidrug resistant pathogens and the depletion of antibiotic options, alternative methods are urgently needed to combat these deadly microorganisms [16,17]. One such method is the use of engineered probiotics, which has several advantages including increased specificity, regulated release of antimicrobial

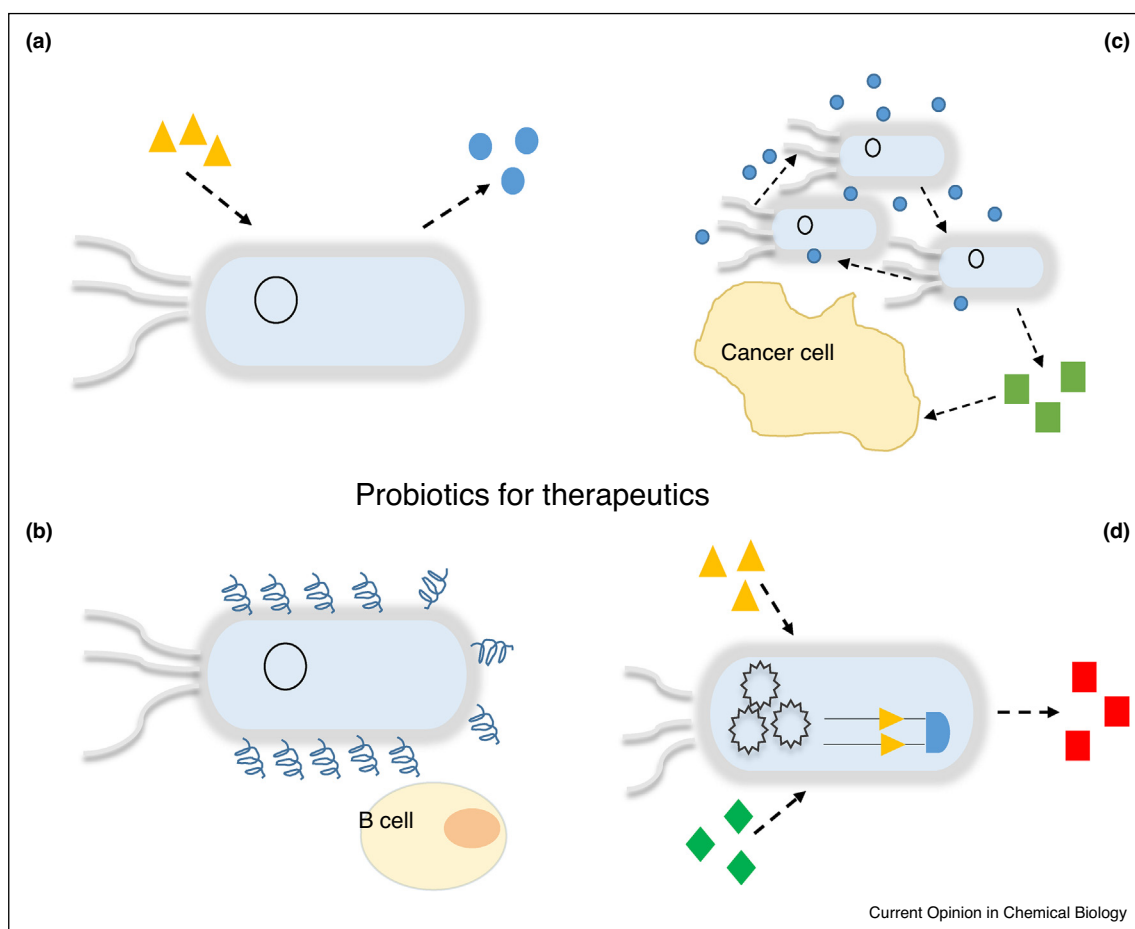
agents and decreased risk of antibiotic resistance development.

The idea of engineering *E. coli* as a sense-and-kill system for planktonic *Pseudomonas aeruginosa* was first described by Saeidi *et al.* [18], with improved specificity and efficiency in degradation of the mature biofilm matrix reported in another study [19]. Similar ideas are recently adopted in *L. lactis* engineered to fight multidrug resistant *Enterococcus* spp. [20,21]. The engineered *L. lactis* was designed to sense *E. faecalis* pheromone cCF10 and in response, release bacteriocins that can kill the multidrug resistant *E. faecalis* [20] (Figure 2a). In another study, the same group engineered *L. lactis* to produce bacteriocins under the control of a chloride-induced promoter that is activated in the gut environment with the aim to target

E. faecium which does not release the cCF10 pheromone [21].

Another pathogen that has been targeted by engineered probiotics is *Clostridium difficile*, which is a leading cause of antibiotic-associated diarrhoea in developed countries. While fecal microbiota transplant (FMT) is currently the most effective treatment for *C. difficile* infection, engineered *L. lactis* that expressed non-toxic fragments of TcdA and TcdB, the virulence factors of *C. difficile*, was observed to elicit an immunity against this pathogen *in vivo*, offering another possible treatment option [22] (Figure 2b). In a separate study, Andersen *et al.* used *L. paracasei* expressing single domain antibodies against TcdB to target *C. difficile* [23]. Similar engineering has also been conducted in probiotics to express relevant antigens

Figure 2



A schematic of strategies to combat health problems with engineered probiotics. **(a)** Probiotics are engineered to sense certain molecules released by pathogens (orange triangles) and in response, release the required therapeutic agents for treating infections. **(b)** Probiotics can be engineered to deliver antigens to treat and prevent allergies in asthmatics. **(c)** Engineered probiotics can produce anti-tumor factors (green squares) upon detection of quorum sensing signals (blue circles) from the accumulation of probiotics attached to the tumor tissue. **(d)** Rational design enables probiotics to be equipped with complex circuits to execute more accurate functions. Engineered probiotics can produce the desired therapeutics (red squares) only upon step-wise detection of two molecules (orange triangles and green diamonds).

to target *Vibrio cholera* [24], *Shigella dysenteriae* [25] and *P. aeruginosa* [26].

Deviating from the sense-and-kill systems, other studies have adopted a more prophylactic approach to combat pathogens. In a study by Zhang *et al.*, an anti-*Helicobacter pylori* vaccine was developed by engineering *L. lactis* to produce *H. pylori* lipoprotein Lpp20 [27]. This engineered probiotic was found to elicit an immune response when tested *in vivo*. In another similar study, modified *Bacillus subtilis* spores were engineered to express *H. pylori* urease B protein on the surface [28]. These recombinant spores were found to generate a humoral response in mice and significantly reduced the *H. pylori* load. *B. subtilis* spores were also engineered to express the antigens of *Mycobacterium tuberculosis* for the development of a vaccine against tuberculosis [29].

Apart from bacterial pathogens, vaccines against viruses such as HIV have also been developed using engineered probiotics [30,31]. Both Kuczkowska *et al.* and Chamcha *et al.* expressed an HIV Gag antigen on the surface of a lactic acid bacteria and showed that it can elicit an immune response and therefore, can be potentially used as a vaccine against HIV.

Engineering probiotics for treatment of allergy and autoimmune diseases

Genetically engineered lactic acid bacteria have been designed to deliver therapeutic heterologous proteins to the host mucosa for treating inflammatory diseases. By feeding arthritic mouse milk fermented with *L. lactis* overexpressing CFA/I fimbriae, arthritis was significantly ameliorated via CD39⁺ Tregs producing TGF- β and IL-10, which potently suppressed TNF- α production and neutrophil influx into the joints [32].

Other efforts aiming to prevent and treat allergies were implemented in *L. lactis* as well. A majority of asthmatics are allergic to house dust mites (HDM), with up to 80% of HDM-allergy patients exerting positive reaction to Der p2, one of the 30 HDM allergens [33]. By feeding the BALB/c mouse with recombinant *L. lactis* overexpressing Der p2 [34], the development of airway inflammation in the Der p2-sensitized mice was significantly prevented via reduced IgE antibody production and T cell reactivity upon HDM exposure (Figure 2b). In addition, delivery of a hypoallergenic macromolecular derivative of the Der p2 (DM) to the intestinal mucosal surface by engineered *L. lactis* was observed to induce around 50% reduction in IgE reactivity as well as up-regulation of specific IgG2a and a decrease in IL-4 level in the spleen [35]. In another study, five Der p2-derived peptides (DPs) containing major T cell epitopes of Der p2 were overexpressed in *L. lactis* [36]. Compared to DM, these DPs not only fully eliminated IgE-binding capacity but also reduced T cells

reactivity, successfully inhibiting Der p2-induced allergic responses in Der p2-sensitized mice.

Not only are engineered *L. lactis* effective in the treatment of allergies *in vivo*, they have also displayed great potential in treating inflammatory bowel diseases (IBDs). If left untreated, IBDs often develop into cancers of the GIT. One of the earliest synthetic probiotics therapeutics developed for this purpose used *L. lactis* to produce IL-10, which was observed to cause 50% reduction in colitis *in vivo* [6]. It has also been found that delivery of IL-10 cDNA from *L. lactis* to host cells is able to confer protective effects against inflammation [37]. *L. lactis* can also be programmed to produce other anti-inflammatory cytokines such as IL-27 [38], thymic stromal lymphopoietin (TSLP) [39] or heme oxygenase-1 (HO-1) [40] to control acute inflammation of the GIT. Apart from *L. lactis*, engineering *B. longum* to express anti-inflammatory peptide alpha-melanocyte-stimulating hormone (α -MSH) was observed to differentially regulate various cytokines, thus ameliorating dextran sulphate sodium (DSS)-induced colitis *in vivo* [41].

Engineering probiotics for targeting and killing tumors

In 2012, a World Health Organization (WHO) project on cancer GLOBOCAN, estimated a total of 14 million new cases of cancer, and the figures are expected to increase by 70% in the next 20 years [42]. On top of this, conventional anti-cancer therapies are usually unable to achieve a complete cancer remission. In the past decade, synthetic biology has stepped in to engineer microbes that can specifically target tumor cells.

The idea of using bacteria to treat cancer originated in the 19th century [43]. Microbiologists over the years have reported a variety of anaerobes that proliferate preferentially in solid tumors [44], thus encouraging the use of these microbes as carriers for anti-cancer agents. Among the various anaerobes tested, the strains from *Clostridium*, *Bifidobacterium*, *Salmonella* and *Escherichia* genera are the most commonly engineered. Facultative anaerobes are especially of interest due to their relative ease in manipulation.

B. longum shows excellent colonization ability in solid tumors [45] and is able to elicit a strong immune response in the host [46]. Making use of its preference to proliferate in hypoxic regions, *B. longum* was recently engineered to express tumstatin, an angiostatin that inhibits proliferation and induces apoptosis in vascular endothelial cells within tumors [47]. Wei *et al.* reported a significant reduction in tumor metastasis, as well as an increase in survival rate in the engineered *B. longum* treated mice compared to the control group. Two groups recently reported success with engineered *B. longum* that expressed herpes simplex virus thymidine kinase

Table 1

List of notable engineered probiotics for the prevention and treatment of human diseases

Disease	Vehicle	Mechanism	Result	Reference
Type I Diabetes	<i>L. gasseri</i>	Reprogramming intestinal epithelial cells (IECs) into insulin-secreting β -cells by GLP-1 (1–37) expressing <i>L. gasseri</i> (LG)	Blood glucose and insulin level in LG-treated rats were not significantly different from nondiabetic control rats, while diabetic rats fed with wild-type (WT) <i>L. gasseri</i> had significantly higher blood glucose and lower insulin level than the control rats	[7*]
Hypertension	<i>L. plantarum</i> NC8	Expressing ACEIP coding sequences from TFP and YFP joined by an arginine linker	Decreased systolic blood pressure in rats during treatment, with effect lasting 10 days after last dose. Other antihypertensive effects observed include increased levels of nitric oxide (NO), and decreased levels of endothelin (ET) and angiotensin II (AngII) in plasma, heart and kidney	[9]
Obesity	<i>EcN</i>	Expressing acylphosphatidylethanolamines (NAPEs) in <i>EcN</i> (pNAPE- <i>EcN</i>)	pNAPE- <i>EcN</i> fed mice gained less body weight, accumulated less fat mass, maintained lower plasma leptin and insulin levels, when compared to the untreated mice	[13*]
Hepatic steatosis	<i>EcN</i>	Expressing fructose dehydrogenase (fdh) or mannitol-2-dehydrogenase (mtlK)	Treated mice showed reduced lipid peroxidation, increase in serum and hepatic antioxidant enzyme activities, as well as restoration of liver injury marker enzymes	[14]
Salmonellosis and Cholera infection	<i>Salmonella</i>	Expression of <i>Vibrio cholera</i> toxin antigen subunit-B heterologous antigen (CtxB) in <i>Salmonella typhimurium</i> Z234-vaccine strain	Increased mucosal response, pro-inflammatory cytokine production as well as decreased <i>V. cholerae</i> colonization was observed in treated mice	[24]
<i>H. pylori</i> infection	<i>B. subtilis</i>	Displaying <i>H. pylori</i> urease B protein on <i>B. subtilis</i> spore coat	<i>H. pylori</i> infected mice that were orally immunized with the engineered <i>B. subtilis</i> showed up to 84% reduction of stomach bacterial load	[28]
HIV infection	<i>L. lactis</i>	Expression of <i>Streptococcus pyogenes</i> T3 pilus fused to an HIV antigen gag P24 on the surface of <i>L. lactis</i> (LL-Gag)	LL-gag immunized mice displayed 3 fold higher CD8 T cell responses in small intestine, as well as activates dendritic cells in Peyer's patches	[31]
Inflammatory bowel diseases	<i>L. lactis</i>	Delivery of IL-10 protein using stress-inducible controlled expression system, and also delivery of IL-10 cDNA cassette into host cells	Mice treated with engineered <i>L. lactis</i> carrying the dual expression systems showed significantly lower weight loss, lower damage scores and higher expression of anti-inflammatory cytokines	[37]
Cancer	<i>Salmonella typhimurium</i>	Expression of interferon-gamma (IFN-) fused to N-terminal region of SipB for secretion from <i>Salmonella</i>	Up to 50% of melanoma cells were killed when infected with the engineered <i>Salmonella</i> . 80% of the mice bearing subcutaneous B16F10 melanoma tumors survived after treatment.	[51]
Cancer	<i>Salmonella typhimurium</i>	Quorum sensing controlled lysis, resulting in release of anti-cancer	80% of tumor-bearing mice which received combined therapy of the engineered <i>Salmonella</i> with conventional chemotherapy drug 5-fluorouracil survived	[54*]
Cancer	<i>Salmonella typhimurium</i>	Engineering a <i>sifA</i> mutant <i>Salmonella</i> that lyses in response to tetracycline and releases cell cycle arrest agents	60% of the <i>Salmonella</i> -infected MCF7 breast cancer cells remained in subG1 population, while only 28% were in G0/G1	[55**]
Cancer	<i>EcN</i>	Expression of glucose and ribose sugars receptor Trz1, which upon activation, triggers expression of GRP reporter	The amount of GFP expressed per bacterium correlated with tumor cell viability	[60*]

(HSV-TK), which in combination with ganciclovir (GCV) is one of the well-studied cancer suicide gene therapies [48,49]. When the engineered microbes were co-administered with GCV, protective effects against human colonic, liver, gastric and breast tumor xenografts in mice were observed.

Another commonly engineered facultative anaerobe for anti-cancer therapeutics is *Salmonella* spp., although some strains of *Salmonella* are well known to cause typhoid fever or food poisoning. The therapeutic applications of engineered attenuated *Salmonella* is multifaceted, which includes engineering *Salmonella* as a drug delivery vehicle by expressing apoptosis related genes such as Fas-associated death domain (*fadd*) [50], cytokines such as interferon-gamma (IFN- γ) [51] to kill tumor cells directly, flagellin B (FlaB) from *Vibrio vulnificus* to boost immune response against tumor cells [52^{••}], or the more complicated circuit-controlled release of anti-cancer agents [53,54[•],55^{••}]. Engineered *Salmonella* may also be used as a tumor detection tool [56]. *Salmonella* has also been developed to target tumors that are difficult to reach or have poor prognosis, such as gliomas [57] and neuroblastomas [58].

Swofford *et al.* demonstrated the use of a quorum sensing (QS) switch to control drug expression in *Salmonella* [53] (Figure 2c), while Din *et al.* took a step further with the QS switch and coupled it with a bacteriophage lysis protein to synchronize bacterial lysis cycles for drug

delivery [54[•]] (Figure 2d). In a different approach, Camacho *et al.* created an elaborate three-part circuit in *Salmonella* in which the *sifA* gene of *Salmonella* responsible for maintaining the *Salmonella*-containing vacuole (SCV) within the host cell is replaced with a set of salicylate-regulated elements [55^{••}]. This allowed the engineered microbes to express anti-tumor agents when induced by salicylate, leading to the release of the microbes into the cytoplasm of the host cell within hours of infection. The third part of this circuit involved the inducible autolysis system, which is activated only in the presence of tetracycline or its analogue anhydrotetracycline (AHT). The group demonstrated the necessity of temporal separation of protein production from protein release for effective killing of the tumor cells.

Other than *Salmonella*, engineered *EcN* also demonstrated the ability to function as cancer diagnostic tools. Engineered *EcN* have been developed to selectively colonize liver tumor in mice and to release a small molecule in the urine that is indicative of liver metastasis [59]. *E. coli* can also be engineered to sense glucose and ribose sugar in solid tumor cell masses [60[•]].

Future directions

The combined understanding and application of synthetic biology on microbes has opened up a vast array of possibilities in therapeutics design (Table 1). With the recent emergence of concepts surrounding gut microbiome and immune diseases, therapeutic probiotics

Table 2

List of the issues of the conventional therapies that can potentially be overcome by engineered probiotics. Noteworthy comparison parameters are marked by †. The advantages of the engineered probiotics marked by * are discussed in the text

Disease (conventional therapy)	Parameter	Conventional therapy	Engineered probiotics
Metabolic Disorders (Drug administration, surgery, lifestyle modification)	Cost of therapy	High, such as bariatric surgery	Low, depending on the method of delivery
	Compliance with the treatment	Low	High
	Duration of the therapeutic effect	Short-term	Long-term*
	Invasiveness	High	Low, only oral intake required
	Others†	Low efficacy	Prevention possible*
Infectious Diseases (antibiotics)	Side effects	Diarrhea, nephrotoxicity	None
	Target pathogen	Only bacteria	Bacteria and viruses*
	Others†	Antibiotic resistance, fewer novel antibiotics	Specificity and regulation possible*
Allergy (Drug administration, immunotherapy)	Side effects	Drowsiness, nausea, diarrhea	None/few
	Compliance with the treatment	Low	High
Cancer (Chemotherapy, immunotherapy, radiation)	Side effects	Nausea, low blood count, increased risk of infection	None
	Specificity	No, damage to normal tissue too	Yes*
	Others†	Incomplete remission, poor efficacy and risk of secondary cancer	Prevention and regulation possible*

targeting multiple sclerosis *in vivo* has become a reality [61]. Engineering of microbes for therapeutic purposes has become increasingly sophisticated over the years, with inducible circuits precisely controlling the expression and release of therapeutic molecules. Advancements in synthetic biology have also improved the efficiency of these microbes as drug delivery vehicles, reducing the number of cells required to elicit therapeutic effects [62].

Although great efforts were made to improve the specificity of these therapeutic microbes in their activity, one main concern is the biocontainment of these engineered probiotics. As these modified probiotics are not naturally occurring, biocontainment strategies should be included in the rational design to reduce dissemination of these microbes into the environment. Recently, *E. coli* capable of coding synthetic amino acids have been developed as an effort towards biocontainment [63,64] but the feasibility has yet to be demonstrated in commonly engineered probiotics strains.

Other than the issue of biocontainment, interactions between synthetic therapeutic probiotics and the commensal microbes in the human body remain poorly understood. Although significant results are observed in various *in vivo* models, similar effects may not be observed in humans as the human enteric microbiome is far more complex than that of animal models. It was recently reported that incorporation of synthetic genetic circuits in non-pathogenic *E. coli* [65^{*}] or human commensal bacterium *Bacteroides thetaiotaomicron* [66] had displayed remarkable ability in sensing and reporting environmental cues, thus allowing greater understanding of the complex relationship between commensal microbiota and host. Nonetheless, these engineered microbes display various benefits over conventional treatments (Table 2). With the recent advancements in synthetic therapeutic probiotics as discussed here, the availability of a new generation of therapeutics is greatly anticipated in the near future.

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- of special interest
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